

OpenCV

- What is OpenCV?
 - Installing OpenCV:
 - Installation via pip
 - Installation via Anaconda (conda install)
 - OpenCV Architecture
 - Understanding OpenCV Ecosystem:
 - `cv2` Module
 - Working with OpenCV and Python
 - OpenCV with Jupyter Notebooks
 - Introduction to Image Processing and Computer Vision
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2. Basic Image Handling in OpenCV

2.1 Image Input/Output

- Reading and Writing Images (`cv2.imread()`, `cv2.imwrite()`)
- Displaying Images (`cv2.imshow()`)
- Accessing Image Properties (Shape, Size, Channels)
- Working with Different Image Formats (JPEG, PNG, TIFF, BMP)

2.2 Image Color Models

- Understanding Color Models (RGB, BGR, Grayscale, HSV, Lab)
- Converting Between Color Spaces (`cv2.cvtColor()`)

2.3 Image Operations

- Basic Image Arithmetic (Addition, Subtraction, Multiplication)
 - Accessing and Modifying Pixel Values
 - Image Resizing and Scaling (`cv2.resize()`)
 - Cropping and Region of Interest (ROI) Selection
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3. Image Transformation and Geometric Operations

3.1 Geometric Transformations

- Translation, Scaling, and Rotation (`cv2.getAffineTransform()`, `cv2.getPerspectiveTransform()`)
- Affine and Perspective Transformations
- Image Warping and Projective Geometry
- Homography Estimation

3.2 Image Flipping, Rotation, and Resizing

- Flipping Images (`cv2.flip()`)
- Rotating Images by Arbitrary Angles
- Changing Image Resolution and Aspect Ratio

3.3 Warping Images and Mapping

- Applying Transformation Matrices
 - Perspective Warping
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4. Image Filtering and Enhancement

4.1 Image Smoothing and Blurring

- Mean Filtering
- Gaussian Blur (`cv2.GaussianBlur()`)
- Median Filtering (`cv2.medianBlur()`)
- Bilateral Filtering (`cv2.bilateralFilter()`)

4.2 Edge Detection

- Canny Edge Detection (`cv2.Canny()`)
- Sobel and Scharr Derivatives for Edge Detection
- Laplacian Edge Detection

4.3 Image Enhancement Techniques

- Histogram Equalization (`cv2.equalizeHist()`)
 - CLAHE (Contrast Limited Adaptive Histogram Equalization)
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5. Feature Detection and Matching

5.1 Corner Detection

- Harris Corner Detection
- Shi-Tomasi Corner Detection (`cv2.goodFeaturesToTrack()`)

5.2 Feature Descriptors

- SIFT (Scale-Invariant Feature Transform)
- SURF (Speeded-Up Robust Features)
- ORB (Oriented FAST and Rotated BRIEF)
- FAST (Features from Accelerated Segment Test)

5.3 Feature Matching

- Brute Force Matcher (`cv2.BFMatcher()`)
 - FLANN (Fast Library for Approximate Nearest Neighbors)
 - RANSAC Algorithm for Outlier Rejection
 - Homography Estimation Using Feature Matches
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6. Contours and Shape Analysis

6.1 Contour Detection

- Finding Contours (`cv2.findContours()`)
- Drawing Contours (`cv2.drawContours()`)
- Contour Approximation and Polygon Approximation

6.2 Shape Detection

- Detecting and Analyzing Shapes (Circles, Squares, Triangles)
- Shape Matching and Template Matching

6.3 Moments and Properties of Contours

- Computing Moments (`cv2.moments()`)
 - Finding Centroids and Bounding Boxes
 - Contour Hierarchy
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7. Object Detection and Tracking

7.1 Object Detection

- Haar Cascade Classifiers for Object Detection
 - Loading Pretrained Haar Classifiers (`cv2.CascadeClassifier()`)
 - Detecting Faces, Eyes, and Other Objects
 - Training Custom Haar Classifiers

7.2 Object Tracking

- Lucas-Kanade Optical Flow
- Kalman Filter for Object Tracking
- Meanshift and Camshift Tracking Algorithms
- Multi-object Tracking

7.3 Advanced Object Detection

- Template Matching (`cv2.matchTemplate()`)
 - SSD (Single Shot Multibox Detector)
 - YOLO (You Only Look Once) in OpenCV
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8. Image Segmentation

8.1 Thresholding Techniques

- Global Thresholding (`cv2.threshold()`)
- Adaptive Thresholding
- Otsu's Binarization

8.2 Region-based Segmentation

- Region Growing Algorithms
- Watershed Algorithm for Image Segmentation

8.3 Edge-based Segmentation

- Canny Edge Detection for Segmentation
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9. Video Processing and Analysis

9.1 Video Capture and Playback

- Reading Video Files (`cv2.VideoCapture()`)
- Writing Video Files (`cv2.VideoWriter()`)
- Displaying Video Frames
- Real-time Video Capture from Webcam

9.2 Frame Processing

- Frame Extraction from Video
- Frame Rate Control
- Video Stabilization

9.3 Motion Detection and Background Subtraction

- Background Subtraction (MOG2, KNN)
 - Optical Flow for Motion Detection
 - Pedestrian Detection and Tracking
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10. Camera Calibration and 3D Vision

10.1 Camera Calibration

- Understanding Camera Intrinsics and Extrinsics
- Camera Calibration with Chessboard Pattern
- Computing Camera Matrix and Distortion Coefficients

10.2 Stereo Vision

- Epipolar Geometry and Stereo Matching
- Depth Map Generation
- 3D Reconstruction from Stereo Images

10.3 Augmented Reality (AR)

- AR Marker Detection
- Pose Estimation
- Real-time 3D Object Augmentation

11. Machine Learning in OpenCV

11.1 Introduction to OpenCV Machine Learning

- Overview of OpenCV Machine Learning Module (`cv2.ml`)
- Training and Using Classifiers (SVM, k-NN, Decision Trees)
- Model Evaluation Metrics (Accuracy, Precision, Recall)

11.2 Image Classification with ML Algorithms

- Using k-NN for Image Classification
- Training a Support Vector Machine (SVM) Classifier
- Decision Tree Classifiers for Image Recognition

11.3 Handwritten Digit Recognition (MNIST Dataset)

12. Deep Learning with OpenCV

12.1 Introduction to Deep Learning in OpenCV

- Integration of DNN (Deep Neural Networks) Module in OpenCV
- Using Pre-trained Models with OpenCV DNN (`cv2.dnn.readNet()`)
- Running CNNs and Other Architectures in OpenCV

12.2 Object Detection with Pre-trained Deep Learning Models

- Using YOLO with OpenCV
- Object Detection with SSD Models
- Real-time Object Detection using Deep Learning

12.3 Face Detection and Recognition with Deep Learning

- Using Pretrained Deep Learning Models for Face Detection and Recognition
 - Implementing Face Recognition Systems
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13. Real-World Applications with OpenCV

13.1 Face Detection and Recognition

- Real-time Face Detection
- Face Recognition using Eigenfaces, Fisherfaces, and LBPH
- Emotion Recognition from Facial Expressions

13.2 Optical Character Recognition (OCR)

- Using Tesseract with OpenCV for OCR
- Text Detection and Extraction from Images

13.3 Hand Gesture Recognition

- Detecting Hand Movements for Interaction
 - Real-time Gesture Recognition with Hand Tracking
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14. Optimization and Performance in OpenCV

14.1 Performance Tuning

- Optimizing OpenCV Code for Speed
- Using Multi-threading with OpenCV
- Using OpenCV with CUDA for GPU Acceleration

14.2 Memory Management

- Efficient Memory Usage in Image and Video Processing
 - Handling Large Datasets
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15. Projects and Case Studies

Beginner Projects

- Image Processing Pipeline for Basic Effects (Blur, Sharpen, Grayscale)
- Object Detection and Tracking with Haar Cascades
- Real-time Face Detection using Webcam

Intermediate Projects

- Build a Camera Calibration System with Chessboard Images
- OCR System for Scanning Documents
- Vehicle Detection and Tracking from Video Feed

Advanced Projects

- Build a Gesture Recognition System
- Implement YOLO for Real-time Object Detection in Video
- Autonomous Car Simulation (Lane Detection, Traffic Sign Recognition)